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One Idea, Two Languages

Module 9 Week B — Property Graphs, Cypher, And NL-Grounded Graph QA

صندوق كفاءات المستقبل
Future Skills Fund

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Istidama consulting

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Get to know Lana – Fun Fact

I spoke at AI Con USA this week

- Opportunity for sharing my work and improving speaking skills
- Great networking
- Check out TechWell conferences–expensive in-person; virtual attendance is free
- Submitting talks and speaking at expensive conferences can get you a ticket + valuable network connections

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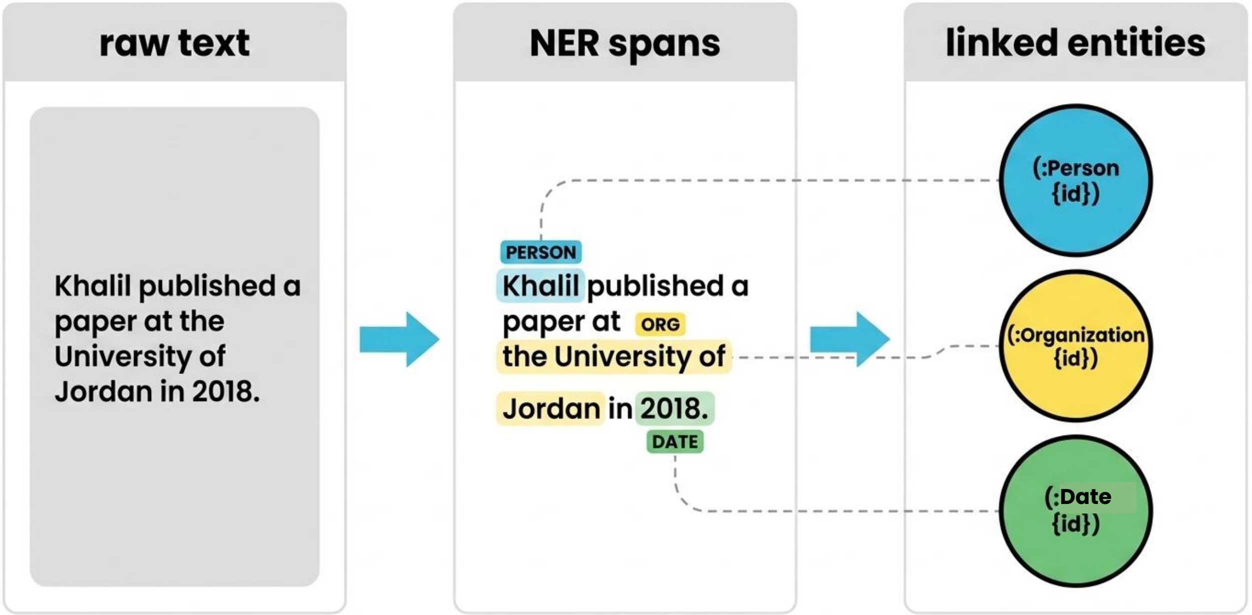
Which Khalil ?

Identity Is The Goal



identity = the deliverable

Two Stacks, One Idea

Standards stack	Application stack
RDF	Property graph
SPARQL	Cypher
Fuseki	Neo4j
Wikidata, biomedical, government, FAIR data	LLM apps, recommendation, fraud, social

Both model data as graphs of nodes and edges

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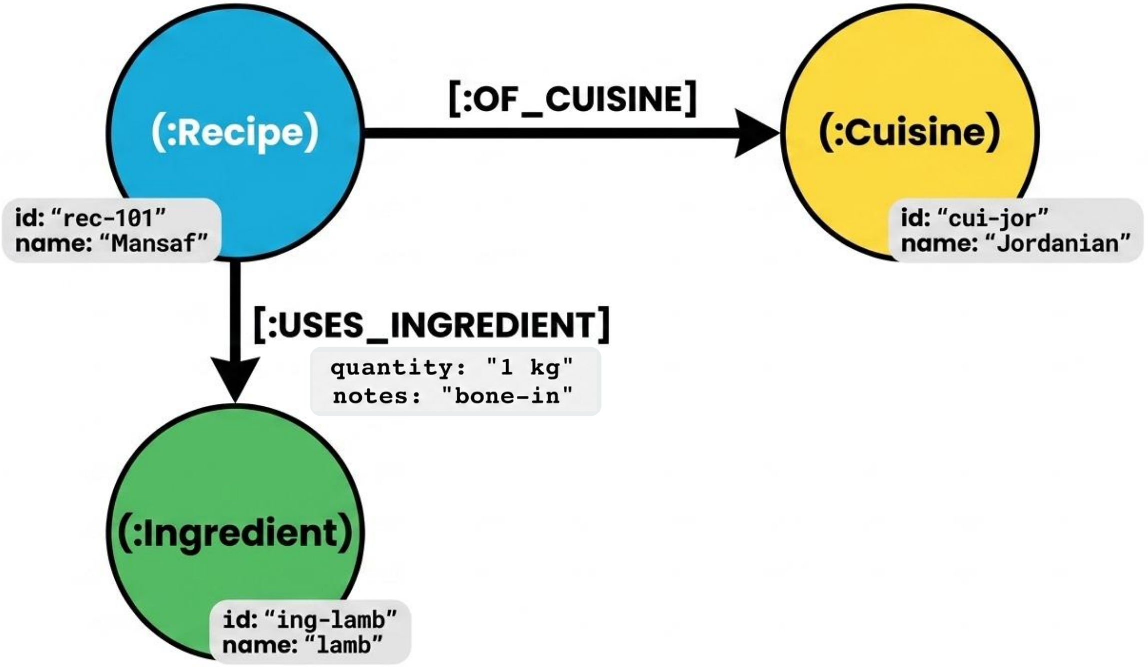
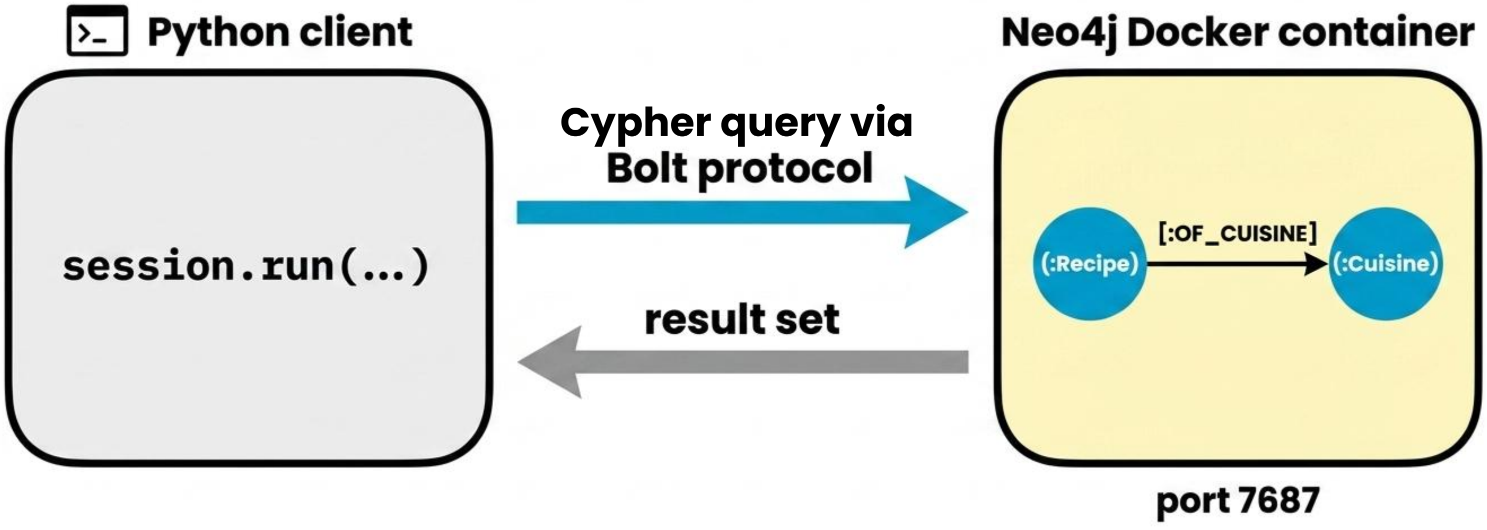
SPARQL And Cypher Side By Side

	RDF / SPARQL (last week)	Property graph / Cypher (this week)
1	?s ?p ?o	(s)-[:REL]->(o)
2	WHERE { ?s rdfs:subClassOf* ?super }	MATCH (s)-[:SUBCLASS_OF*0..]->(super)
3	FILTER (?o = "...")	WHERE o.prop = "..."
4	OPTIONAL { ... }	OPTIONAL MATCH (...)
5	Fuseki SPARQL endpoint	Neo4j Bolt endpoint

Nodes, Relationships, Properties



Neo4j And The Bolt Endpoint

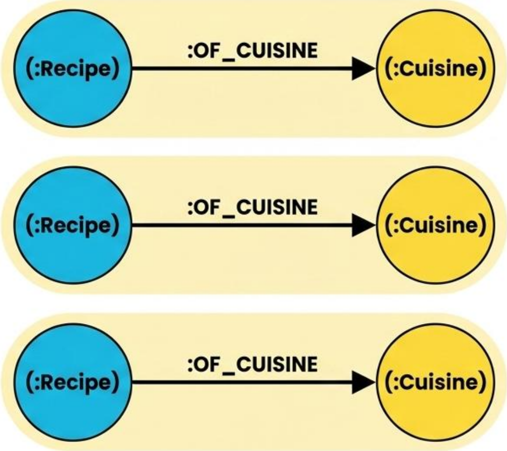


Cypher Is Pattern Matching (same)

the query

```
MATCH (r:Recipe)-[:OF_CUISINE]->(c:Cuisine)
RETURN r, c
```

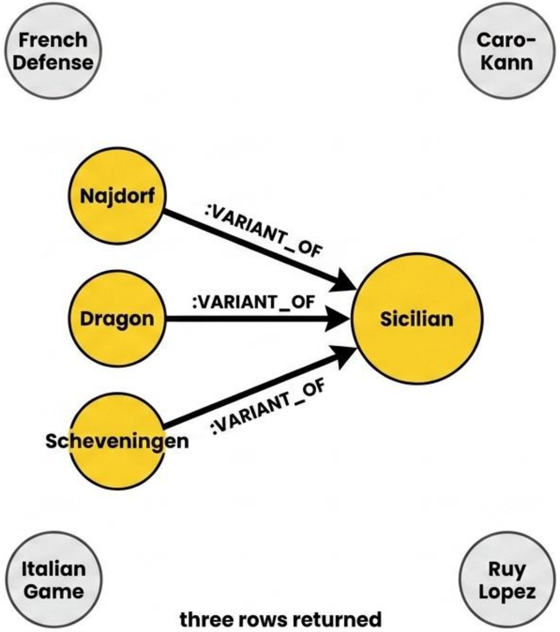
the matches



three matches = three rows

One Hop 🐰

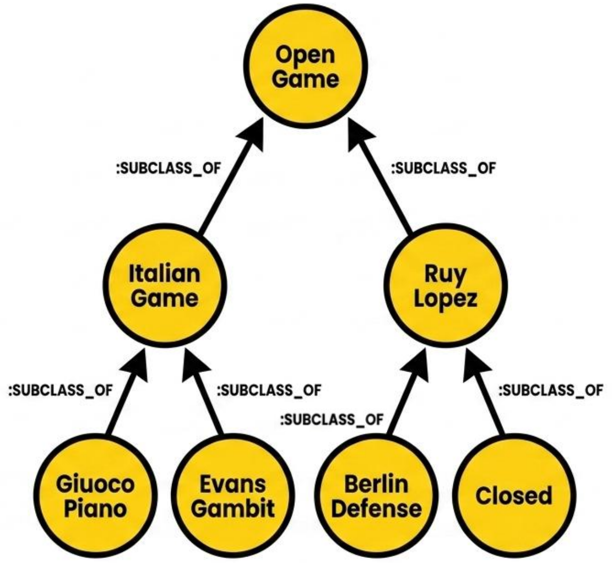
MATCH (o:Opening)-[:VARIANT_OF]->(p: Opening {name: 'Sicilian'}) RETURN o.name



Many Hops

```
MATCH (o:Opening)-[:SUBCLASS_OF*0..]-
  ->(:Opening {name: 'Open Game'})
RETURN o.name
```

zero or more hops



seven rows returned

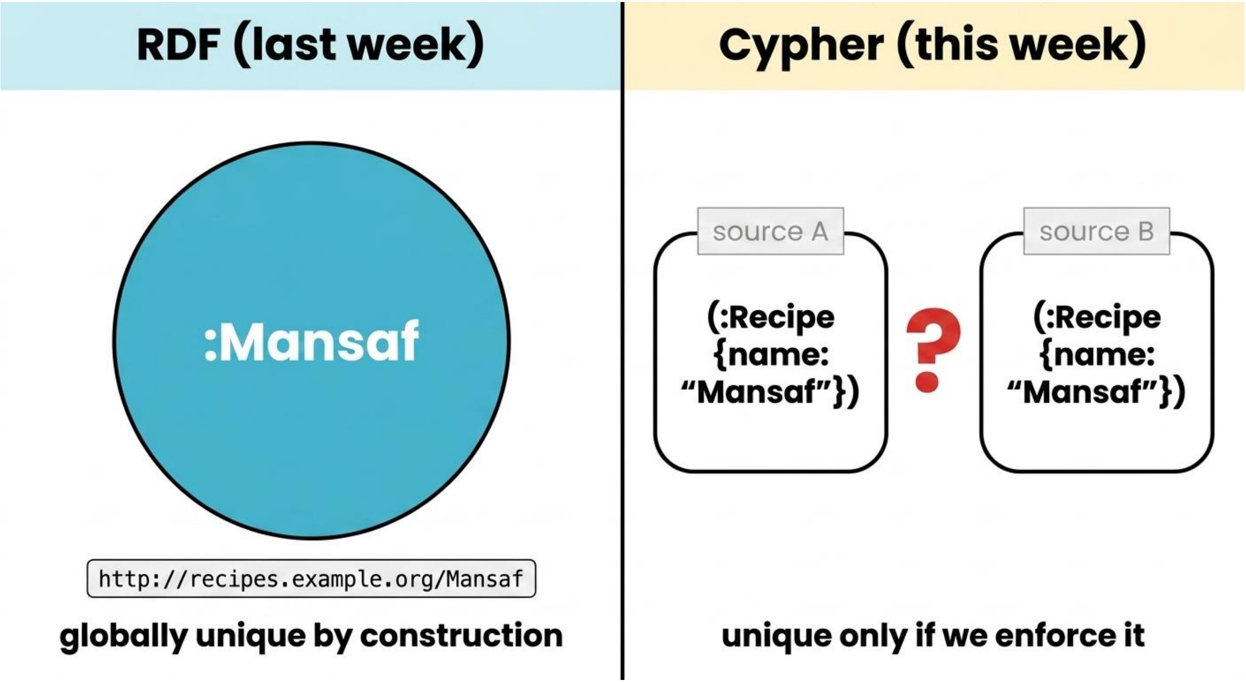
Use Dollar-Sign Parameters

safe ✅	silent bug 💀

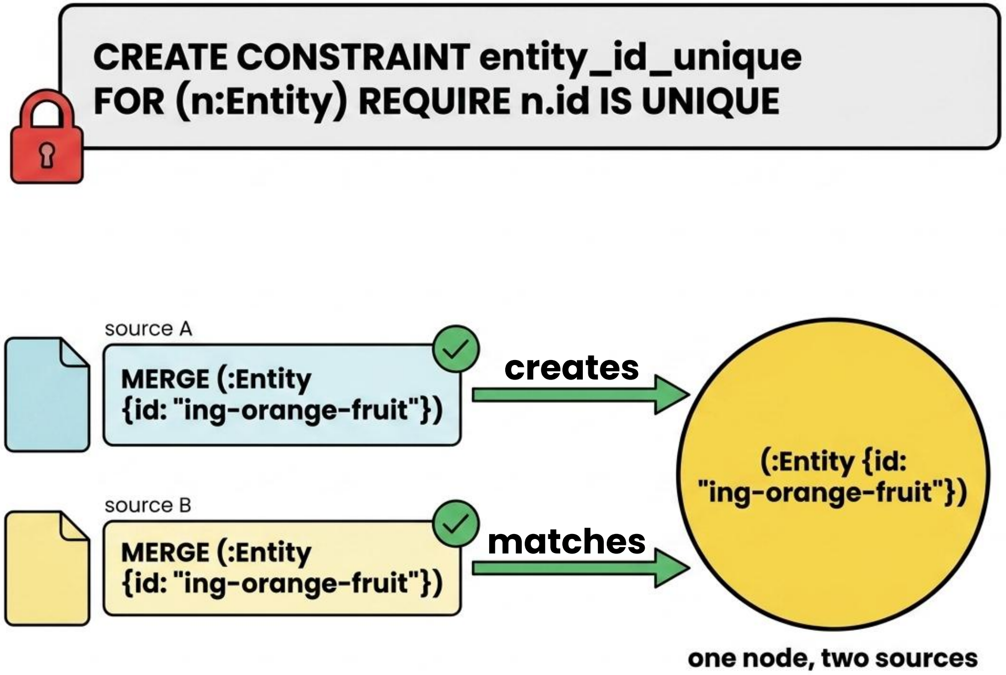
`MATCH (o:Opening {name: $opening_name}) RETURN o`
 	
 `session.run(query, opening_name='Sicilian')`
	`"MATCH (o:Opening {name: '" + user_input + "'}) RETURN o"`
	
	 untrusted concat

same shape, very different safety

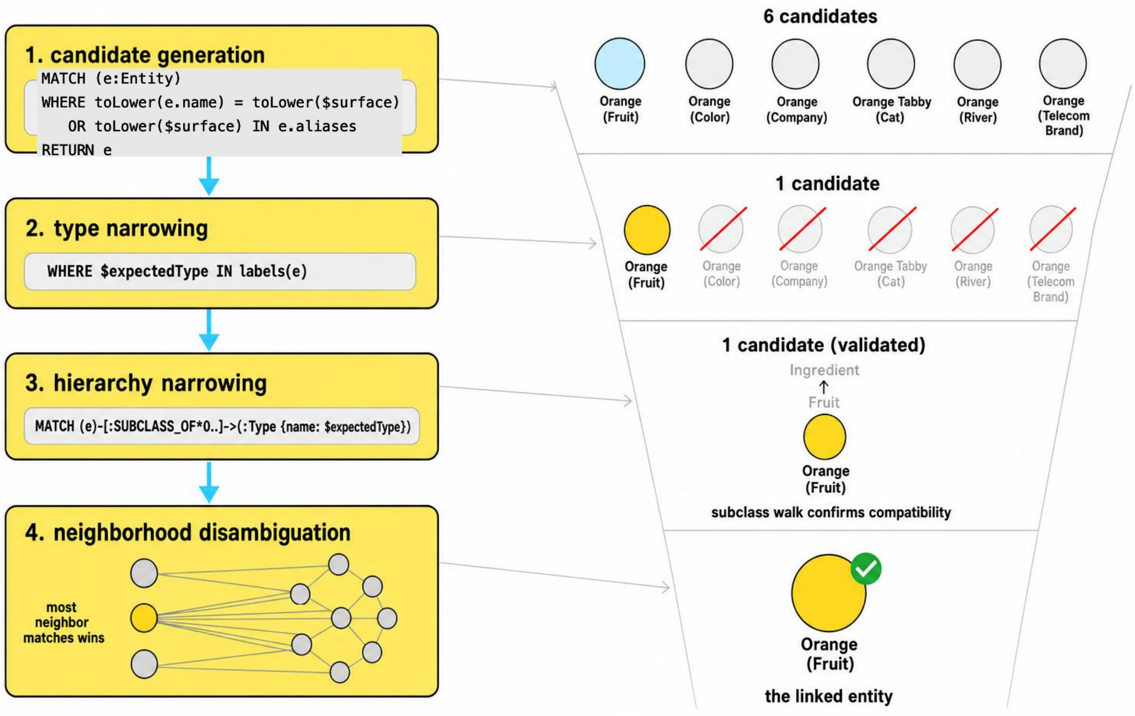
Review: Automatic Identity with RDF



The Uniqueness Constraint



The Linker



1. candidate generation
MATCH (e:Entity)
WHERE toLower(e.name) = toLower($surface)
OR toLower($surface) IN e.aliases
RETURN e
2. type narrowing
WHERE $expectedType IN labels(e)
3. hierarchy narrowing
MATCH (e)-[:SUBCLASS_OF*0..]->(:Type {name: $expectedType})
4. neighborhood disambiguation
most neighbor matches wins
6 candidates
Orange (Fruit)
Orange (Color)
Orange (Company)
Orange Tabby (Cat)
Orange (River)
Orange (Telecom Brand)
1 candidate
Orange (Fruit)
Orange (Color)
Orange (Company)
Orange Tabby (Cat)
Orange (River)
Orange (Telecom Brand)
1 candidate (validated)
Ingredient
↑
Fruit
Orange (Fruit)
subclass walk confirms compatibility
Orange (Fruit)
the linked entity

Precision, Recall, F1

entity linking

	predicted: linked	predicted: NIL
gold: linked	**TP** (true positive)	**FN** (false negative)
gold: NIL	**FP** (false positive)	**TN** (true negative)

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

F1 = 2PR / (P + R)

Wrong entity ID affects both FP & FN.

FP for the predicted entity.

FN for the missed gold entity.

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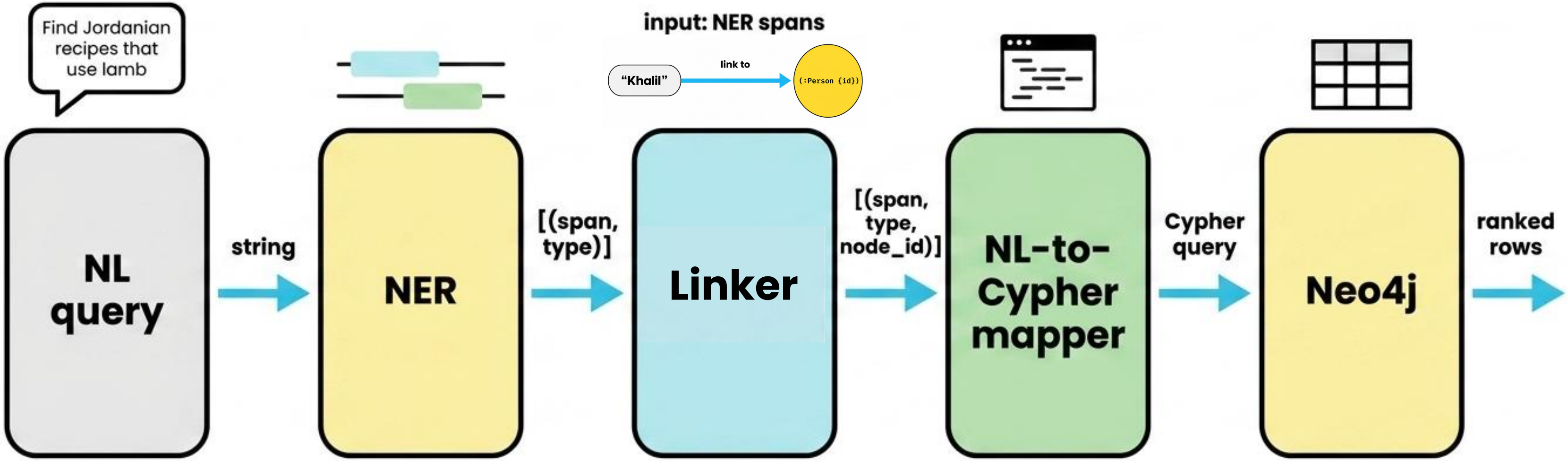
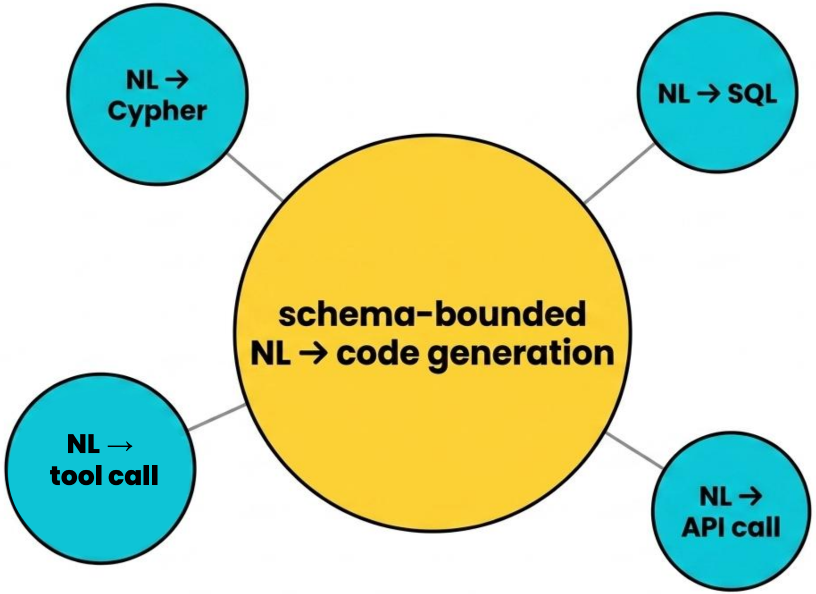
From Text To Cypher



Natural Language To Cypher



bounded schema = template-able OR LLM-tractable

What The Small Model Got Wrong

the question

Find Jordanian recipes that use lamb

the small model produced ☠️

```
MATCH (c:Cuisine {name: 'Jordanian'})-[:SUBCLASS_OF]->(r:Recipe)
WHERE c.name = 'lamb'
RETURN r
```

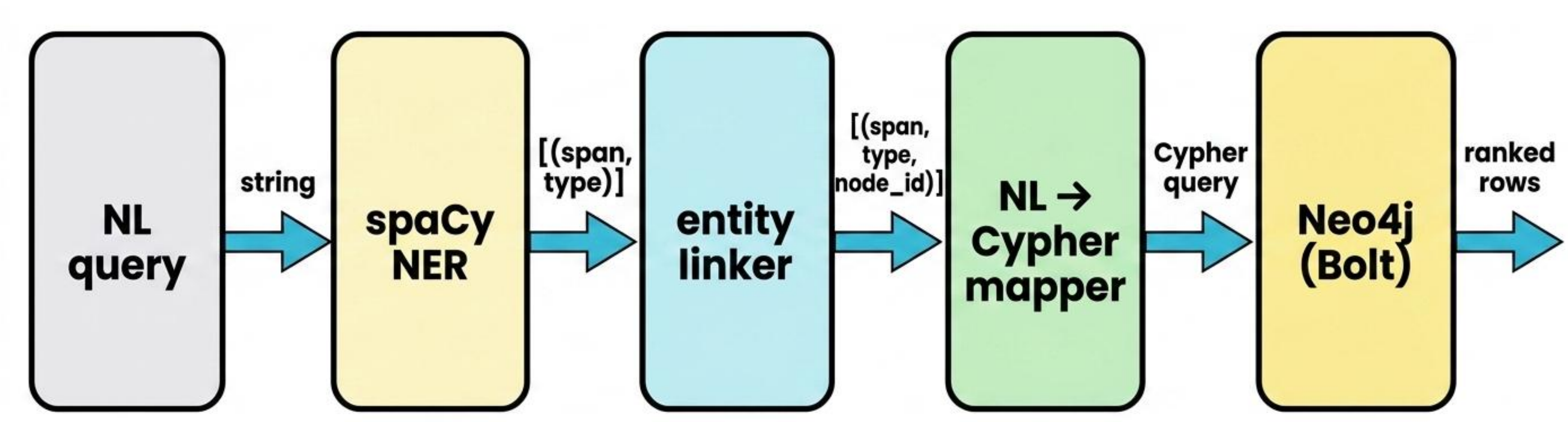
wrong relationship

wrong node

the deterministic mapper emitted ✅

```
MATCH (r:Recipe)-[:OF_CUISINE]->(:Cuisine {name: 'Jordanian'})
MATCH (r)-[:USES_INGREDIENT]->(:Ingredient {name: 'lamb'})
RETURN r
```

The Integration



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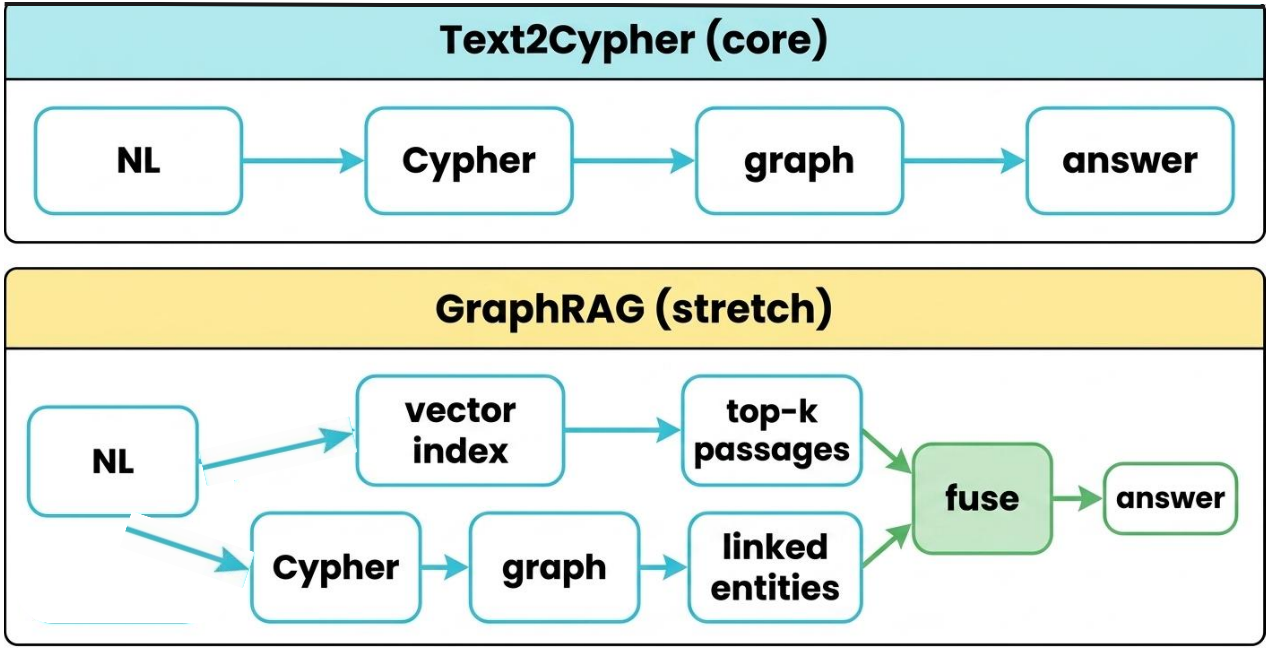
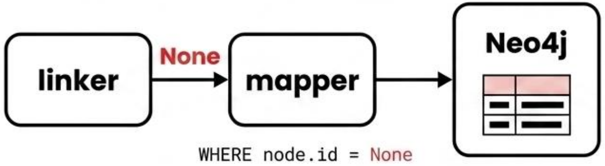
Text2Cypher Vs GraphRAG



M8 stack + M9 stack, combined at answer time

Two Silent-Failure Risks

NIL propagation



wrong answer, looks right

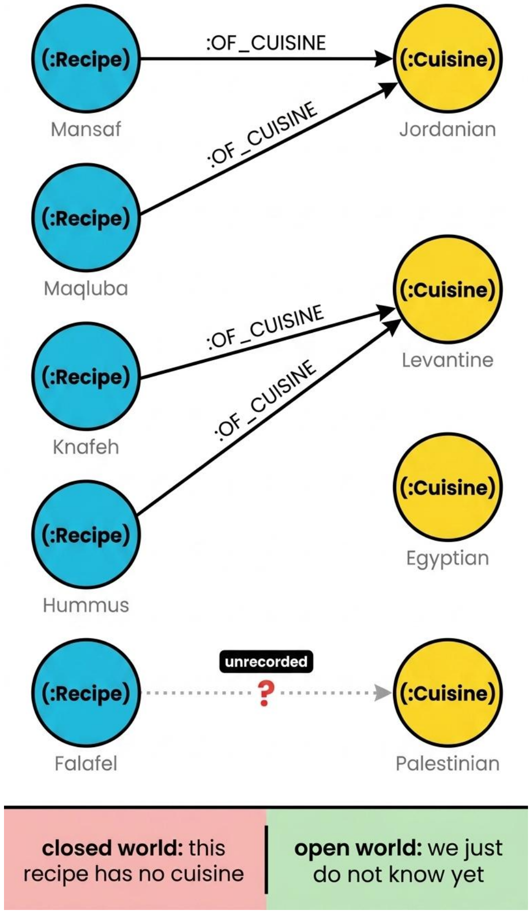
f-string injection

```
f"MATCH (o {name: '{user_input}'}) RETURN o"
```

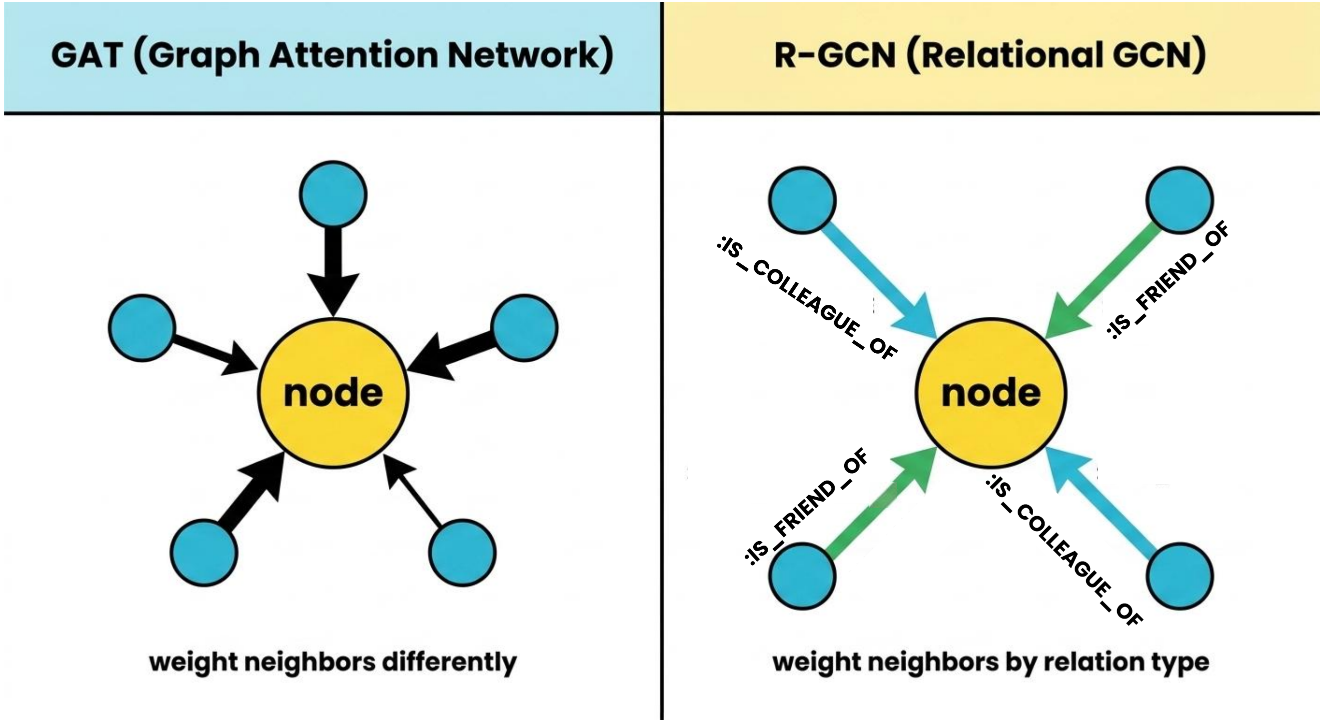
untrusted

parses, executes, returns the wrong rows

The Graph Is Incomplete



GNNs — Embeddings From Neighbors



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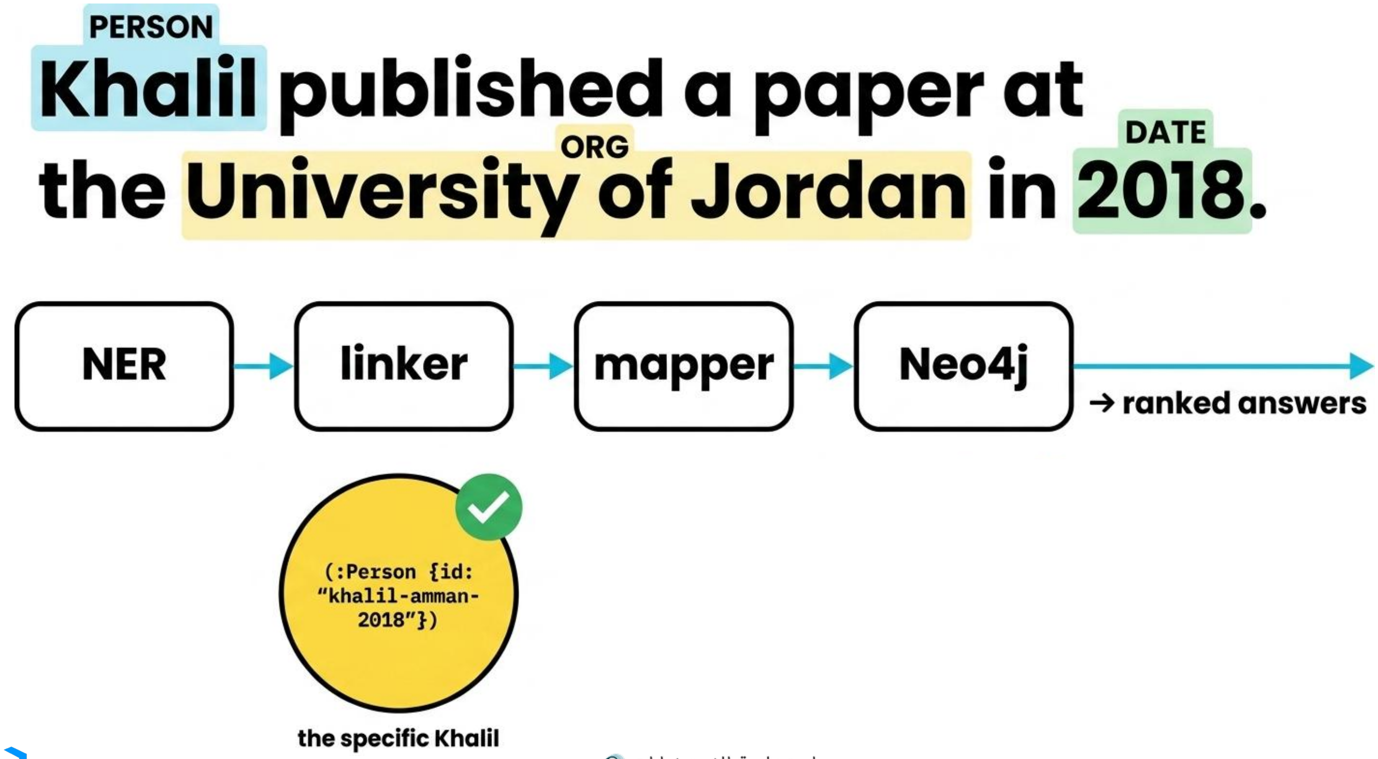
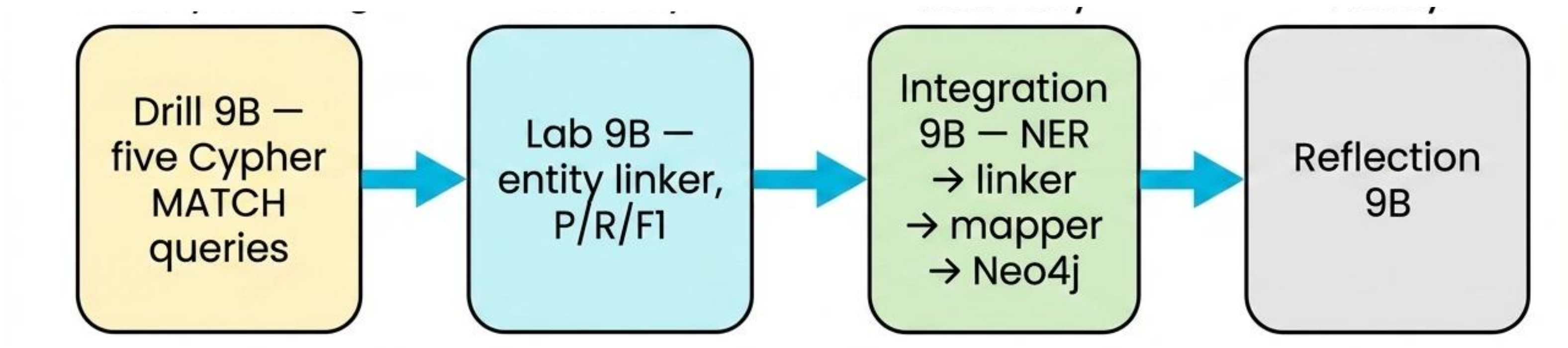
Back To The Sentence



Your Week


Drill 9B — five Cypher MATCH queries
Lab 9B — entity linker, P/R/F1
Integration 9B — NER → linker → mapper → Neo4j
Reflection 9B

Questions